Project Machine Learning
Milestone 2: Ablation Studies and Multi-Task Evaluation of Swin Transformer

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January 11, 2026

1 Introduction

In Milestone 1, we successfully reimplemented Swin Transformer Tiny from scratch and validated its correctness through perfect pretrained weight transfer and linear probing on CIFAR-100. Milestone 2 builds upon this foundation by conducting comprehensive ablation studies of Swin's key design decisions, including shifted windows, relative position bias, and window size. Additionally, we systematically compare Swin Transformer against ResNet and ViT backbones across three core vision tasks: image classification on ImageNet-1K, object detection on COCO, and semantic segmentation on ADE20K. This report analyzes how each architectural choice contributes to performance and efficiency, providing empirical evidence for Swin's superiority in hierarchical vision modeling.

2 Ablation Studies

All ablation studies are conducted on a subset of ImageNet-1K consisting of 50,000 training images and 5,000 validation images across 1,000 classes. This setup isolates the effect of each architectural design choice while keeping compute requirements manageable.

The training configuration is identical across all experiments, images are resized to 224×224 using bicubic interpolation, a batch size of 128 is used, and optimization is performed with AdamW at a base learning rate of 5×10^{-4} (linear scaling rule). The learning rate follows a cosine decay schedule after a 3-epoch linear warmup. Training is run for 20 epochs to enable rapid iteration and to capture differences in early convergence speed and final performance. Validation is performed using Top-1 accuracy on the validation set.

Results are reported after 20 epochs, as this duration is sufficient to reveal meaningful trends in learning dynamics and representational capacity.

2.1 Ablation Study: Shifted Window Mechanism

To investigate the importance of the shifted window mechanism (SW-MSA), we trained two identical Swin-Tiny models from scratch on ImageNet-1K for 20 epochs: one with the default alternating W-MSA/SW-MSA pattern and one with regular W-MSA only in every block (shifted windows disabled).

Configuration	**Top-1 Acc (%)**	**Δ vs Baseline**
Swin-Tiny (baseline, shifted windows)	7.58	—
Swin-Tiny (no shifted windows)	5.96	**–1.62 pp**

Table 1: Impact of disabling shifted windows after 20 epochs of training from scratch on ImageNet-1K.

Disabling shifted windows results in a substantial performance drop of **1.62 percentage points** (7.58 % → 5.96 % Top-1 accuracy). This confirms the original paper's finding that the shifted window mechanism is critical for establishing cross-window connections and enabling effective long-range modeling, even in the early stages of training.

The loss curves show that the model without shifted windows converges more slowly and reaches a higher final validation loss, indicating reduced representational capacity. This ablation demonstrates that the alternating shioft design is a key contributor to Swin Transformer's strong transfer learning ability.

2.2 Ablation Study: Relative Position Bias

To assess the contribution of the relative position bias mechanism, we trained two Swin-Tiny models from scratch on ImageNet-1K for 20 epochs: one with relative position bias enabled (default) and one with the bias table set to zero.

Configuration	Top-1 Acc (%)	Δ vs Baseline
Swin-Tiny (baseline, relative bias)	7.58	—
Swin-Tiny (no relative bias)	6.36	**–1.22 pp**

Table 2: Impact of disabling relative position bias after 20 epochs of training from scratch on ImageNet-1K.

Disabling relative position bias results in a **1.22 percentage point** drop in Top-1 accuracy (7.58 % $\rightarrow$ 6.36 %). This substantial degradation confirms the original paper's finding that relative position bias is essential for encoding spatial relationships within windows and providing translation invariance — a key inductive bias for vision transformers. The loss curves further illustrate the effect: the model without bias converges more slowly and reaches a higher final validation loss, indicating reduced learning efficiency and representational capacity.

2.3 Ablation Study: Absolute vs Relative Position Encoding

To evaluate the impact of position encoding strategy, we compared the standard relative position bias (default) with learned absolute positional embeddings (ViT-style) on Swin-Tiny trained from scratch on ImageNet-1K for 20 epochs.

Configuration	Top-1 Acc (%)	Δ vs Baseline
Swin-Tiny (baseline, relative bias)	7.58	—
Swin-Tiny (absolute pos embed)	8.60	**+1.02 pp**

Table 3: Impact of using absolute positional embeddings instead of relative bias after 20 epochs of training from scratch on ImageNet-1K.

Surprisingly, replacing relative bias with absolute positional embeddings improved Top-1 accuracy by **1.02 percentage points** (7.58 % $\rightarrow$ 8.60 %). This result contrasts with the original Swin V1 paper's finding that relative bias outperforms absolute embeddings, suggesting that on from-scratch training with limited epochs, absolute embeddings may provide stronger positional priors or better optimization stability. The loss curves show that the absolute embedding variant converges faster in early epochs and achieves lower final validation loss, indicating improved training dynamics despite the theoretical advantages of relative bias for translation equivariance.

2.4 Ablation Study: Window Size

To evaluate the impact of window size on performance, we trained three Swin-Tiny models from scratch on ImageNet-1K for 20 epochs with window sizes M = 4, 7 (default), and 14. Depth was adjusted to keep the total number of parameters approximately constant.

Window Size (M)	Top-1 Acc (%)	Δ vs Default (M=7)
7 (baseline)	7.58	—
4	4.68	**–2.90 pp**
14	8.02	**+0.44 pp**

Table 4: Impact of window size after 20 epochs of training from scratch on ImageNet-1K.

The results show that the default window size M=7 is not optimal for our short-training setting. M=14 yields a **0.44 percentage point** improvement (7.58 % → 8.02 %), while M=4 causes a severe drop of **2.90 pp** (7.58 % → 4.68 %). This contrasts with the original Swin V1 paper, where M=7 was optimal on longer training. Our findings suggest that larger windows (M=14) provide better long-range modeling in early training stages, while smaller windows (M=4) limit receptive fields and hinder convergence. The loss curves confirm faster convergence and lower final validation loss for M=14.

2.4.1 Sanity Check: Training from Scratch on CIFAR-100

To further verify the stability and learning capacity of our reimplemented Swin-Tiny, we trained the model from random initialization on CIFAR-100. After 50 epochs it achieved 61.13 % Top-1 accuracy on the test set, increasing to 65.14 % after 100 epochs. The consistent improvement and smooth convergence confirm that our implementation is fully functional and capable of effective learning even without pretrained weights.

3 From-Scratch Training Comparison

To assess the learning capacity and architectural differences when training from random initialization, we trained three models from scratch on ImageNet-1K for 40 epochs using identical training settings (AdamW, 3-epoch warmup, cosine decay, RandAugment + MixUp + CutMix + Label Smoothing 0.1, batch size 32). The training was performed on a subset of 100,000 training images and 50,000 validation images to make longer training feasible within compute constraints.

Model	Top-1 Acc (%)	Backbone Params
ResNet-50	42.68	25.6M
ViT-B/16	17.28	29.9M
Swin-Tiny (ours)	**26.73**	**28.3M**

Table 5: Top-1 accuracy on ImageNet-1K test set after 40 epochs of training from scratch.

The results show clear architectural differences in training-from-scratch performance: - ResNet-50 achieves the highest accuracy (42.68 %), demonstrating the strong inductive bias of convolutional layers for image classification without pretraining. - Swin-Tiny reaches 26.73 %, significantly outperforming ViT-B/16 (17.28 %) by 9.45 percentage points, confirming the advantage of hierarchical design and local window attention over global self-attention in low-data regimes. - ViT-B/16 shows the weakest performance, consistent with known difficulties of training pure transformers from scratch without large-scale pretraining or heavy regularization. These observations are based on partial training (40 epochs), and the models have not yet fully converged. In the original Swin Transformer paper Liu et al. (2021), full training (300 epochs) shows Swin outperforming both ResNet and ViT variants, suggesting that with sufficient training time, Swin's hierarchical and local attention structure may close the gap to ResNet and maintain its advantage over ViT. The current results should therefore be interpreted as early indicators of learning dynamics rather than final converged performance.

4 Dense Prediction Tasks

4.1 Architectural Adaptations

Multi-stage feature outputs are required by the applied detection and segmentation frameworks, which rely on feature representations at different spatial resolutions and semantic depths. The Swin architecture was therefore modified to expose the outputs of all four hierarchical stages. These stage-wise feature maps follow a structure analogous to that of ResNet-50 He et al. (2016) and provide progressively coarser spatial resolutions with increasing semantic abstraction. Each stage output is normalized using a Layer Normalization layer and returned as a tuple, enabling seamless integration with downstream components such as feature pyramid networks and segmentation decoders.

In addition, the architecture was adapted to support variable input image sizes, which are inherent to datasets such as COCO and ADE20K Lin et al. (2014); Zhou et al. (2017). For COCO, image resolutions vary across samples and are typically padded to a common size within each batch, resulting in batch-wise fixed but globally variable resolutions. In contrast, ADE20K is commonly preprocessed to a fixed input resolution. As the Swin Transformer architecture assumes specific divisibility constraints on the input resolution, padding mechanisms were introduced at multiple points in the network. During patch embedding, input images are padded so that their spatial dimensions are divisible by the patch size. Window-based self-attention further requires the feature map height and width to be divisible by the window size, which requires additional padding when necessary. Finally, during patch merging operations, padding is applied again to ensure even spatial dimensions, allowing neighboring patches to be merged correctly. These modifications enable the Swin Transformer to process images of arbitrary resolution while preserving its hierarchical feature structure.

4.2 Object Detection

Object detection aims to localize and classify multiple objects within an image by predicting bounding boxes and corresponding class labels. Unlike image classification, detection requires both semantic understanding and precise spatial localization. A central challenge arises from the large variation in object scales, which motivates the use of multi-scale feature representations. Detection performance is commonly evaluated using average precision metrics that jointly assess classification and localization quality across multiple overlap thresholds Lin et al. (2014).

4.2.1 Understanding COCO

The COCO (Common Objects in Context) 2017 dataset Lin et al. (2014) consists of 123,287 publicly available annotated images, comprising 118,287 images in the training set and 5,000 images in the validation set. The images are taken from real-world scenes depicting everyday environments with natural object arrangements. Image resolutions vary substantially, with the shorter image side typically ranging from approximately 200 pixels to over 1,000 pixels, while preserving diverse aspect ratios Lin et al. (2014).

Each image contains a variable number of annotated bounding boxes spanning 80 object classes. These classes cover common categories such as people, vehicles, animals, household objects, and sports equipment. The average number of object instances per image is approximately 7, and bounding boxes may overlap within a single image due to object occlusions and crowded scenes. In addition to bounding box annotations for object detection, the dataset also provides pixel-level instance segmentation masks for each object instance. The presence of occlusion, background clutter, and large variations in object scale makes COCO particularly challenging for visual recognition.

Evaluation on COCO is performed using Average Precision (AP) metrics computed over multiple Intersection over Union (IoU) thresholds, where IoU measures the overlap between a predicted object region and the corresponding ground-truth annotation. The primary metric, AP, averages over IoU thresholds ranging from 0.50 to 0.95 in increments of 0.05, while AP50 and AP75 report performance at fixed IoU thresholds of 0.50 and 0.75, respectively, reflecting increasing localization accuracy requirements Lin et al. (2014). Due to its complexity and comprehensive annotation scheme, COCO has become the de facto benchmark for object detection and instance segmentation, and numerous state-of-the-art methods are developed and evaluated using it He et al. (2017).

4.2.2 Cascade Mask R-CNN Pipeline

Cascade Mask R-CNN is used as the detection and instance segmentation framework, as it achieved the highest overall bounding-box average precision, including AP, AP50, and AP75, among the evaluated methods in Liu et al. (2021).

The method relies on a multi-scale feature representation, as objects in natural images appear at varying spatial resolutions and require features with different levels of semantic abstraction. The backbone network produces feature maps at multiple stages, where earlier stages preserve higher spatial resolution while later stages capture stronger semantic information at coarser resolution. These stage-wise outputs are unified by a Feature Pyramid Network (FPN), which serves as the neck of the architecture Lin et al. (2017). FPN constructs a feature pyramid with consistent channel dimensionality across scales by combining lateral connections with a top-down pathway. This design enables high-level semantic information to be propagated to higher-resolution feature maps while maintaining fine-grained spatial detail, which is essential for robust multi-scale object detection.

Based on the resulting feature pyramid, a Region Proposal Network (RPN) generates a set of candidate object regions Ren et al. (2016). Anchors with predefined scales and aspect ratios are placed densely across all spatial locations of each pyramid level. During training, anchors are assigned to ground-truth objects using an IoU–based matching strategy, allowing the network to distinguish foreground regions from background and to learn coarse object localization. The RPN is optimized jointly for objectness classification and bounding-box regression, producing a ranked set of region proposals after non-maximum suppression. A fixed number of top-scoring proposals per image are retained and forwarded to the subsequent Region of Interest (RoI) processing stage.

The Cascade RoI head refines region proposals through a sequence of classification and bounding-box regression stages Cai and Vasconcelos (2018). At each stage, RoI Align is applied to extract fixed-size feature representations from the appropriate pyramid level without quantizing RoI coordinates, thereby preserving precise spatial alignment between image regions and feature maps Cai and Vasconcelos (2019). RoIs are assigned to pyramid levels based on their spatial scale to ensure compatibility with the corresponding feature-map stride. Each cascade stage is trained with progressively increasing IoU thresholds for assigning positive samples, enforcing increasingly strict localization requirements. In addition, later stages regress smaller residuals relative to the incoming proposals, encouraging finer-grained localization refinements and mitigating the mismatch between training and inference IoU distributions, particularly at higher IoU thresholds.

In addition to bounding-box prediction, a mask head is attached to the RoI features to perform instance segmentation Cai and Vasconcelos (2019). The mask branch predicts a binary segmentation mask for each positive RoI at a fixed spatial resolution and is trained using a per-pixel classification objective. Although primarily introduced for instance segmentation, this auxiliary task provides dense, spatially precise supervision that encourages the network to learn shape-aware and localization-sensitive features. As a result, the mask branch can indirectly benefit detection performance.

4.2.3 Training Setup

Training was conducted following the protocol described in Liu et al. (2021). All experiments were implemented using the MMDetection framework Chen et al. (2019). Multi-scale training was employed, where input images are resized such that the shorter side is randomly sampled between 480 and 800 pixels while the longer side is constrained to a maximum of 1333 pixels. Optimization was performed using the AdamW optimizer with an initial learning rate of 0.0001, a weight decay of 0.05, and a total batch size of 16.

In contrast to the original training setup, which employs a 3× training schedule, models were trained using a 1× schedule (12 epochs) due to computational constraints. For backbone initialization, pretrained weights from the `timm` library Wightman (2021) were used for both Swin-T and ResNet-50.

4.2.4 Results

Backbone	$\mathbf{AP}^{\text{box}}$	$\mathbf{AP}^{\text{box}}_{50}$	$\mathbf{AP}^{\text{box}}_{75}$	$\mathbf{AP}^{\text{mask}}$	$\mathbf{AP}^{\text{mask}}_{50}$	$\mathbf{AP}^{\text{mask}}_{75}$	**params**
ResNet-50	42.6	60.5	46.3	37.5	57.8	40.1	82M
Swin-Tiny	**47.9**	**66.9**	**51.9**	**41.5**	**64.2**	**44.7**	86M

Table 6: Object detection and instance segmentation with Cascade Mask R-CNN on COCO val2017 (1× schedule, `timm`-pretrained backbone weights).

Results shown in table 6 indicate that Swin-Tiny outperforms ResNet-50 for both object detection and instance segmentation, even when trained with a reduced 1× schedule. Specifically, Swin-Tiny achieves an improvement of 5.3 AP for bounding-box detection and 4.0 AP for instance segmentation compared to ResNet-50. This suggests that Swin-Tiny provides more expressive and semantically rich multi-stage feature representations for dense prediction tasks than the convolutional ResNet backbone.

When compared to the results reported in Liu et al. (2021), both backbones exhibit lower performance due to the shorter training schedule. The ResNet-50 backbone shows a drop of 3.7 AP in object detection relative to the 3× schedule, while Swin-Tiny shows a smaller drop of 2.6 AP. This observation indicates that ResNet-50 benefits more from extended training, whereas Swin-Tiny converges faster and requires less adaptation by the detection heads to achieve strong performance.

The trends observed for AP50 and AP75 are consistent with the overall AP results, with Swin-Tiny outperforming ResNet-50 across all reported metrics. No single IoU threshold exhibits an anomalously large discrepancy, suggesting that the performance gains of Swin-Tiny are robust across different localization strictness levels and align well with the findings reported in Liu et al. (2021).

Despite the overall performance improvements, both backbones struggle with small objects, which is consistent with known challenges in the COCO benchmark Lin et al. (2014). Swin-Tiny achieves an AP of 30.8 for small objects in bounding-box detection, compared to 25.5 for ResNet-50. While Swin-Tiny demonstrates a clear advantage, the relatively low absolute performance indicates that small-object detection remains challenging and may benefit from longer training schedules.

4.3 Semantic Segmentation

4.3.1 Understanding ADE20K

The ADE20K dataset Zhou et al. (2017) is a large-scale benchmark for semantic segmentation and scene parsing, covering 150 semantic categories including objects, stuff classes, and object parts. The dataset contains 25,210 images in total, with 20,210 for training, 2,000 for validation, and 3,000 for testing Zhou et al. (2019). Images are collected from diverse indoor and outdoor scenes with exhaustive pixel-level annotations, averaging approximately 19.5 object instances per image.

A key characteristic of ADE20K is its open-vocabulary annotation strategy, resulting in a long-tailed class distribution where some categories appear frequently (e.g., wall, floor, sky) while others occur rarely. This imbalance, combined with the diversity of scene types, makes ADE20K particularly challenging for segmentation models.

Evaluation is performed using mean Intersection over Union (mIoU), which computes IoU between predicted and ground-truth masks for each class and averages across all 150 categories. Pixel accuracy is reported as a secondary metric. Due to its diversity and challenging class distribution, ADE20K has become the standard benchmark for semantic segmentation research Xiao et al. (2018).

4.3.2 UPerNet Pipeline

UPerNet (Unified Perceptual Parsing Network) is used as the segmentation framework, as it achieved strong performance across multiple backbones in the original Swin Transformer evaluation Liu et al. (2021). The architecture combines two complementary modules for multi-scale feature aggregation: a Pyramid Pooling Module (PPM) and a Feature Pyramid Network (FPN) Xiao et al. (2018).

The backbone network produces feature maps at four hierarchical stages with progressively decreasing spatial resolution and increasing channel dimensionality. Table 7 summarizes the multi-scale feature configurations for each backbone. ResNet-101 and DeiT-S produce analogous multi-scale representations, though with different channel configurations.

Feature	**Resolution**	**Swin-T**	**ResNet-101**	**DeiT-S**
C1	$\frac{H}{4} \times \frac{W}{4}$	96	256	384
C2	$\frac{H}{8} \times \frac{W}{8}$	192	512	384
C3	$\frac{H}{16} \times \frac{W}{16}$	384	1024	384
C4	$\frac{H}{32} \times \frac{W}{32}$	768	2048	384

Table 7: Multi-scale backbone feature configurations (channel dimensions). DeiT-S uses a multi-level neck to create pseudo-hierarchical features from its single-scale output.

The Pyramid Pooling Module Zhao et al. (2017) is applied to the deepest feature map to capture global context at multiple scales. Adaptive average pooling is performed at four spatial resolutions (1×1, 2×2, 3×3, and 6×6), followed by 1×1 convolutions that reduce the channel dimension to 512. Each pooled representation is then upsampled back to the original feature resolution and concatenated with the input, yielding a context-enriched feature map that is further refined through a bottleneck convolution.

The Feature Pyramid Network fuses information across all four backbone stages through lateral connections and a top-down pathway Lin et al. (2017). Each stage output is first projected to a common channel dimension of 512 via 1×1 convolutions. Starting from the deepest level, features are progressively upsampled and added to the corresponding shallower-level representations, enabling high-level semantic information to propagate to finer spatial resolutions. After the top-down fusion, 3×3 convolutions are applied at each level to produce the final multi-scale feature set.

For the final prediction, all FPN outputs are upsampled to a common resolution (one-quarter of the input size) and concatenated, resulting in a 2048-channel feature tensor. A bottleneck convolution reduces this to 512 channels, followed by dropout regularization and a 1×1 classifier that produces per-pixel class logits. The output is then bilinearly upsampled to the original input resolution. This fusion-at-reduced-resolution strategy maintains computational efficiency while preserving spatial detail through the final upsampling step.

For DeiT-S, which produces single-scale features at $\frac{1}{16}$ resolution, a multi-level neck module creates pseudo-hierarchical representations through bilinear interpolation at scales of 4×, 2×, 1×, and 0.5×, following the approach described in SETR Zheng et al. (2021). This enables DeiT to interface with the same UPerNet decoder as the natively hierarchical Swin and ResNet backbones.

4.3.3 Training Setup

Training was conducted following the protocol described in Liu et al. (2021). All experiments were implemented using a custom PyTorch training pipeline. Images are resized to a fixed resolution of 512×512 pixels using bilinear interpolation for images and nearest-neighbor interpolation for segmentation masks to preserve class indices. Data augmentation consists of random horizontal flipping during training, while validation uses deterministic center crops. Input images are normalized using ImageNet mean and standard deviation values.

Optimization is performed using the AdamW optimizer with an initial learning rate of 6×10^{-5} and a weight decay of 0.01. Training proceeds for 160 epochs with a 2-epoch linear warmup phase. A batch size of 8 is used due to GPU memory constraints, with mixed-precision training (bfloat16) enabled for computational efficiency. The loss function is pixel-wise cross-entropy with an ignore index of 255 for unlabeled regions.

For backbone initialization, ImageNet-1K pretrained weights from the `timm` library Wightman (2021) are used for all three backbones. Swin-T uses `swin_tiny_patch4_window7_224`, ResNet-101 uses standard torchvision pretrained weights, and DeiT-S uses `deit_small_patch16_224`. All backbones are fine-tuned jointly with the UperNet decoder, which is randomly initialized.

4.3.4 Results

Backbone	mIoU (%)	Pixel Acc (%)	Params
ResNet-101	40.39	78.95	86M
DeiT-S	41.34	79.66	52M
Swin-Tiny	**44.41**	**80.51**	60M

Table 8: Semantic segmentation on ADE20K validation using UPerNet decoder (160 epochs, ImageNet-1K pretrained backbones).

The results confirm the superiority of Swin Transformer over both convolutional and plain transformer baselines for semantic segmentation. Swin-T achieves 44.41% mIoU, outperforming ResNet-101 by 4.02 percentage points and DeiT-S by 3.07 percentage points, despite having fewer parameters than ResNet-101 (60M vs. 86M). The hierarchical feature structure and shifted window attention mechanism enable Swin-T to capture both local details and global context more effectively than the compared architectures.

DeiT-S reaches 41.34% mIoU, slightly surpassing ResNet-101 while using significantly fewer parameters (52M vs. 86M). However, it exhibits earlier overfitting during training compared to Swin-T, with best validation performance achieved at epoch 43 rather than continuing to improve throughout the full training schedule. This suggests that while the multi-level neck enables DeiT to interface with UPerNet, the lack of native hierarchical features limits its capacity for dense prediction tasks. Further analysis of training dynamics and architectural trade-offs is provided in the Discussion section.

5 Discussion

5.1 Impact of Design Choices

Our ablation studies on Swin-Tiny confirm the importance of two key design elements. Disabling shifted windows reduces accuracy by 1.62 pp, validating their role in establishing cross-window connectivity for long-range modeling. Similarly, removing relative position bias causes a 1.22 pp drop, confirming its value for encoding spatial relationships within windows.

However, two findings diverge from the original paper: absolute positional embeddings outperform relative bias by 1.02 pp, and larger window size M=14 beats the default M=7 by 0.44 pp. These reversals likely reflect our short training schedule (20 epochs vs. 300), where simpler positional priors and larger receptive fields aid early convergence. The original paper shows relative bias and M=7 are optimal at convergence, particularly for dense prediction tasks where translation equivariance matters Liu et al. (2021). This suggests that optimal hyperparameters are training-duration dependent—a practical consideration for resource-constrained settings.

The three evaluation tasks reveal distinct architectural demands. Image classification from scratch favors strong inductive biases—ResNet-50 leads due to built-in locality and translation equivariance, while ViT-B/16 struggles without pretraining. For dense prediction tasks with pretrained backbones, hierarchical structure becomes decisive: Swin-T outperforms ResNet-50 by 5.3 box AP on COCO and ResNet-101 by 4.02 mIoU on ADE20K, despite comparable or fewer parameters. DeiT-S, lacking native hierarchy, requires a multi-level neck to interface with UPerNet and still underperforms Swin-T by 3.07 mIoU while overfitting earlier. These results suggest that for dense prediction, native hierarchical features are more effective than post-hoc interpolation, and that Swin's combination of local attention with cross-window connectivity strikes an effective balance between the inductive bias of convolutions and the flexibility of global attention.

5.2 Swin vs ResNet vs ViT

Our cross-task evaluation reveals a consistent pattern: architectural inductive biases determine performance in data-limited settings, while hierarchical structure is critical for dense prediction tasks.

When training from scratch on ImageNet-1K (40 epochs), ResNet-50 leads with 42.68% accuracy, followed by Swin-T (26.73%) and ViT-B/16 (17.28%). This ordering reflects the strength of built-in priors—convolutions encode locality and translation equivariance directly, while Swin's window attention provides intermediate bias that outperforms ViT's global attention by 9.45 percentage points Liu et al. (2021).

With pretrained backbones, the hierarchy reverses. On COCO object detection with Cascade Mask R-CNN, Swin-T achieves 47.9 box AP, outperforming ResNet-50 (42.6 AP) by 5.3 points. The gap is consistent across metrics: +6.4 AP_{50}, +5.6 AP_{75}, and +4.0 mask AP. On ADE20K semantic segmentation with UPerNet, Swin-T reaches 44.41% mIoU, outperforming ResNet-101 (+4.02 pp) and DeiT-S (+3.07 pp) despite fewer parameters. DeiT's pseudo-hierarchical features via multi-level neck interpolation Zheng et al. (2021) cannot match Swin's natively learned representations, as evidenced by earlier overfitting (epoch 43 vs. continued improvement).

These results validate the core design principles of Swin Transformer: shifted window attention provides efficient cross-window connectivity with linear complexity, while the hierarchical structure enables native multi-scale feature extraction—making it well-suited for general-purpose visual modeling across classification, detection, and segmentation.

5.3 Limitations and Future Work

Computational Constraints. All experiments were conducted on a shared GPU cluster with single-GPU nodes under constrained computational resources. This limited batch sizes to 8 for segmentation (512×512), compared to 16 in the original Swin paper Liu et al. (2021). Training schedules were also shortened: 1× (12 epochs) for COCO detection versus 3× in the paper, and 40 epochs for from-scratch ImageNet comparison versus 300 epochs. These constraints may underestimate the performance ceiling of all architectures, particularly for transformers which benefit from longer training.

Backbone Coverage. Our comparison focuses on Swin-T, ResNet-50/101, and DeiT-S/ViT-B due to memory constraints. Larger variants (Swin-S/B, ResNet-152, ViT-L) and ImageNet-22K pretraining—which yield substantial gains in the original paper—remain unexplored. Additionally, we did not evaluate DeiT on object detection, where its quadratic complexity and lack of native hierarchy pose significant challenges for high-resolution inputs.

Ablation Scope. Ablation studies used 20-epoch training on a 50K-image ImageNet subset, sufficient to reveal trends but insufficient for convergence. Some findings—such as absolute position embeddings outperforming relative bias—may not generalize to longer training schedules where relative bias shows stronger performance on dense prediction tasks Liu et al. (2021). Future work should extend ablations to full training durations.

Future Directions. Given the current infrastructure, practical next steps include extending ablation studies to longer training schedules to verify whether our short-training findings (e.g., absolute vs. relative position bias) hold at convergence. Adding DeiT to object detection experiments would complete the backbone comparison across all three tasks. If additional compute becomes available, training with 3× schedules on COCO and full 300-epoch runs on ImageNet would allow direct comparison with published results.

6 Conclusion

This milestone evaluated the Swin Transformer architecture through ablation studies and multi-task benchmarking, confirming its effectiveness as a general-purpose vision backbone while providing insights into key architectural design choices.

Our ablation studies validated the critical importance of Swin's core design elements. The observed performance reversals likely reflect the shortened training schedule, since simpler positional priors and larger receptive fields facilitate faster early convergence. For object detection and semantic segmentation, Swin Transformer strikes a strong balance between efficiency and expressiveness. Its shifted window mechanism enables linear-complexity cross-window modeling, and its hierarchical design naturally produces multi-scale features well suited for these tasks. Although experiments were limited to shorter training schedules and smaller model variants due to computational constraints, consistent performance gains across all tasks validate Swin's design principles and suitability as a general-purpose vision backbone.

Milestone 3 will build on these results by exploring advanced training strategies, architectural variants, and downstream tasks requiring fine-grained visual understanding.

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